Introduction to Composites Material

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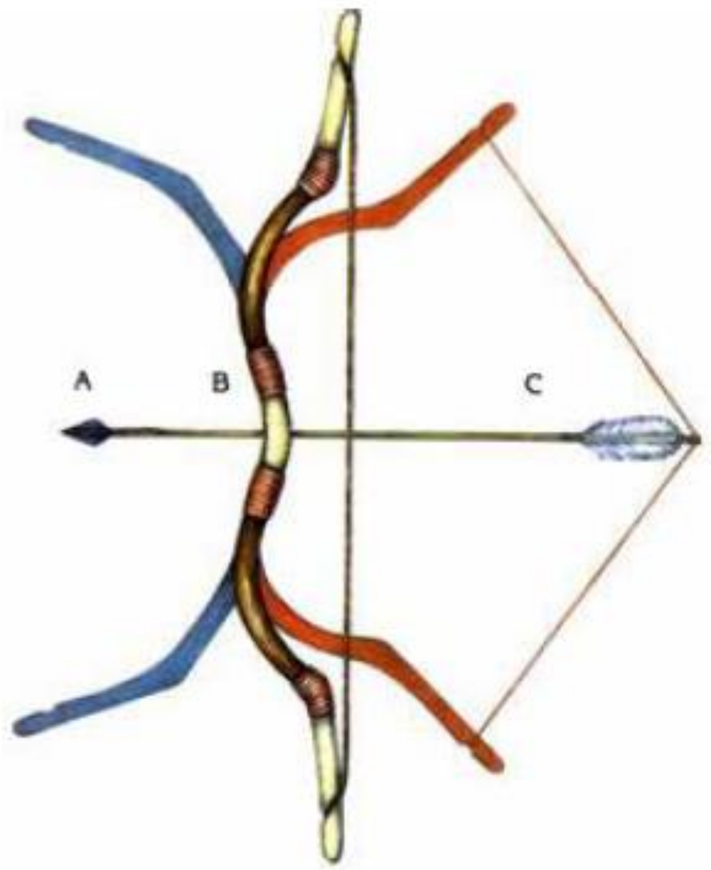
COMPOSITE MATERIALS

Historical significance

Oldest application/existence of composite material
1200 A.D. - Mongols invented the first composite bow



Materials Used:

Wood, Horn, Sinew (Tendon), Leather, Bamboo and Antler (Deer horn)

Horn and Antler: **naturally flexible and resilient**

Sinews: **back tendons or hamstrings of cows and deer**

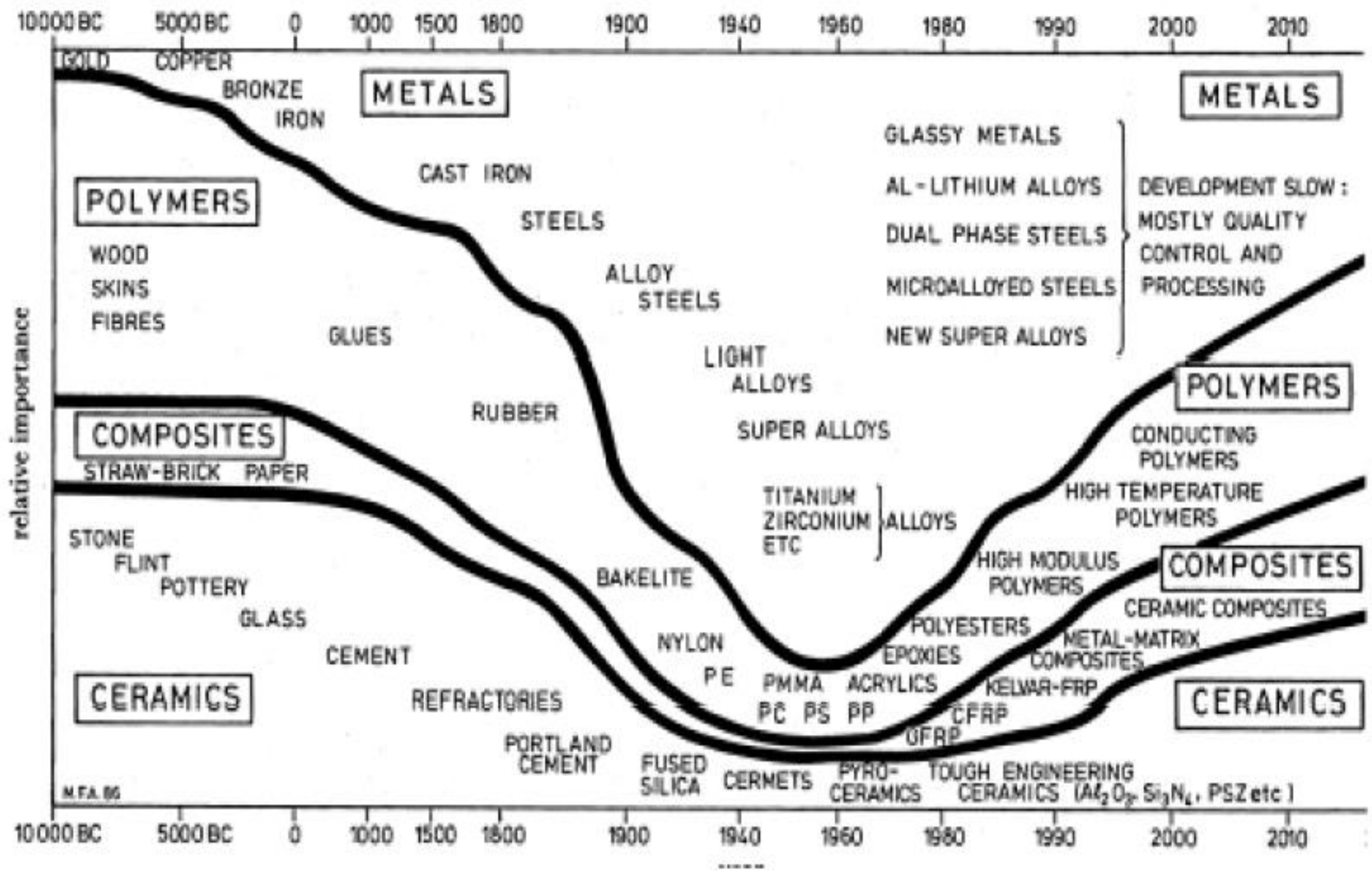
Glue: **From bladder of fish**

Strings: **Sinew, Horse hair, Silk**

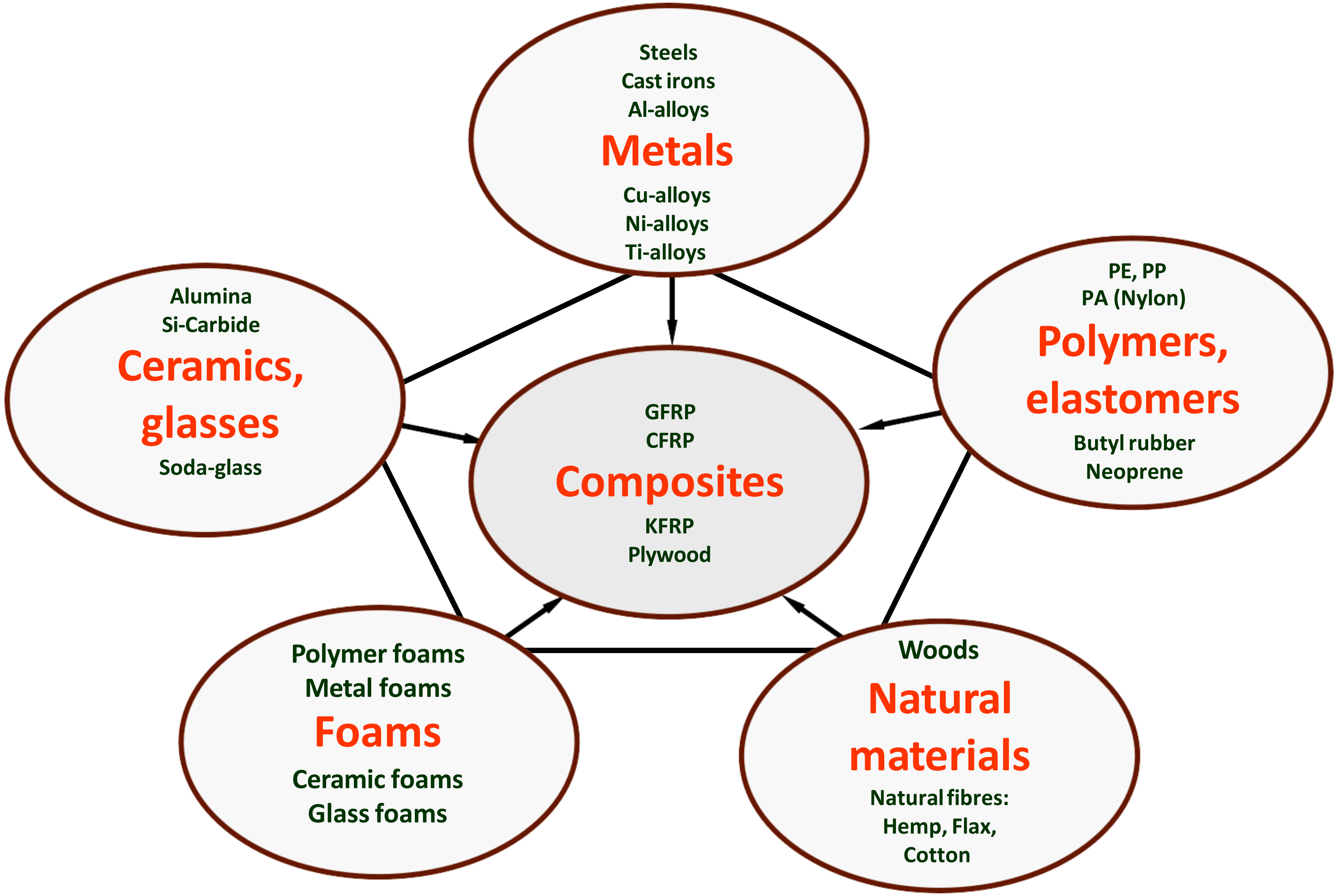
Overall processing time was almost a year !

Composite Bow – dates back to 3000 BC (Angara Dating)

Evolution of Materials



The world of materials



Definition

- A broad definition of composite is: Two or more chemically distinct materials which when combined have improved properties over the individual materials.

- The constituents retain their identities in the composite; that is, they do not dissolve or otherwise merge completely into each other, although they act in concert.

In another words

A composite material is a material that consist of one or more discontinuous components (particles/fibers/reinforcement) that are placed in a continuous medium (matrix)

In a fibre composite the matrix binds together the fibres, transfer loads between the fibres and protect them from the environment and external damage

Components in a Composite Material

Nearly all composite materials consist of two phases:

1. Primary phase - forms the *matrix* within which the secondary phase is imbedded

1. Secondary phase - imbedded phase sometimes referred to as a *reinforcing agent*, because it usually serves to strengthen the composite
   - The reinforcing phase may be in the form of fibers, particles, or various other geometries

Functions of the matrix material (Primary Phase)

- It takes the load and transfer it to the reinforcement

- It binds or holds the reinforcement phase and protects the same from external damage (either chemical and mechanical)

- Matrix also separates the individual fibers and prevents brittle cracks from passing completely across the section of the composite

The Reinforcing Phase (Secondary Phase)

The reinforcing phase is usually of low density, strong, stiff and thermally stable (being metals and ceramics)

The major load on the composite is born by the reinforcing phase

It impedes the motion of dislocations if it is in the form of dispersoids

- Imbedded phase following shapes:
  - Fibers
  - Particles
  - Flakes

Factors in Creating Composites

Matrix material

Reinforcement material

- *Concentration*
- *Size*
- *Shape*
- *Distribution*
- *Orientation*

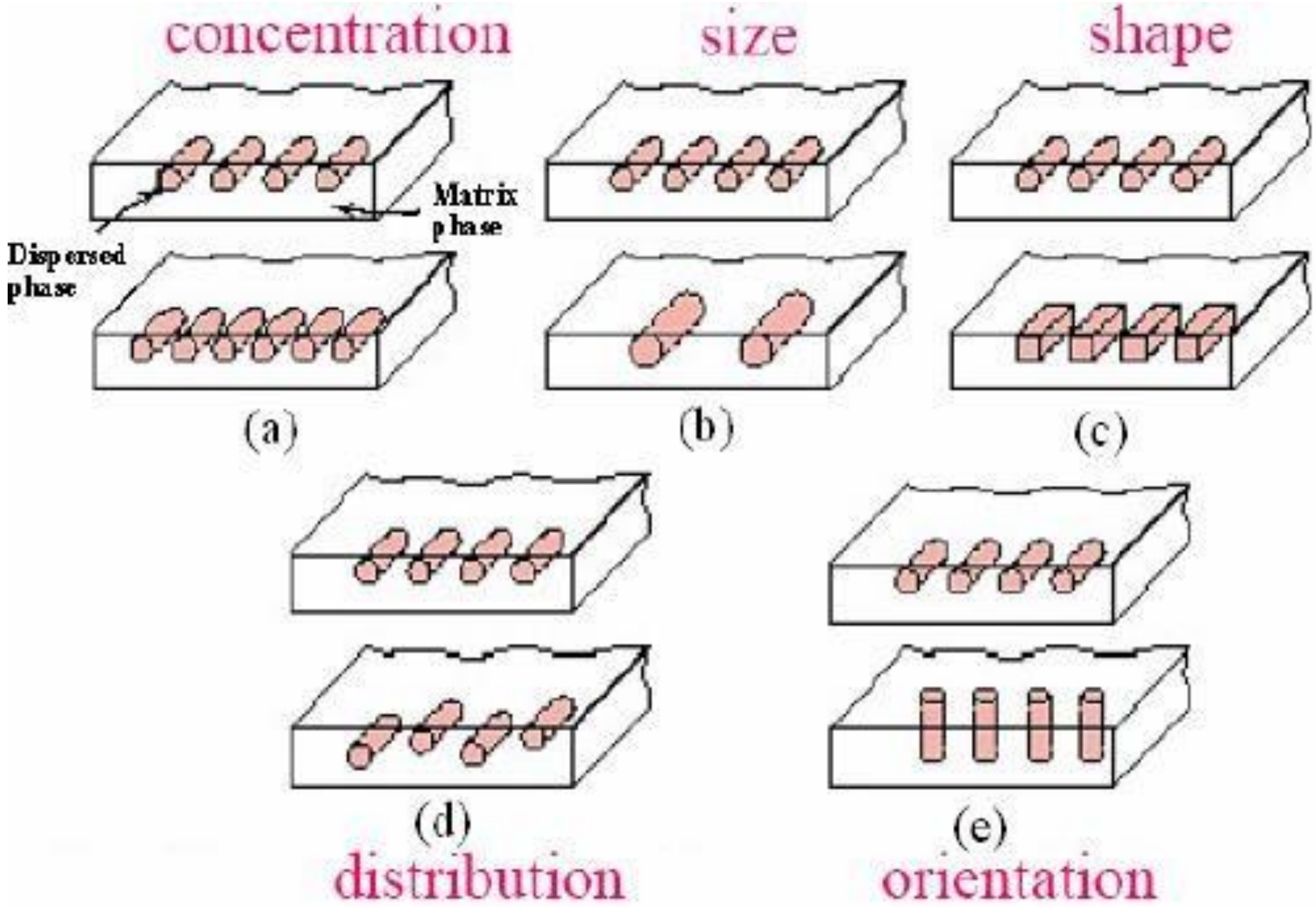


Fig. 1 responsible factors in creating composite material

General requirement of composite

1. **The Second phase (fibers or particles) must be uniformly distributed throughout the matrix and must not be in direct contact with one another**
2. The constituents of the composite should not react with one another at high temperature; otherwise, the interfacial bond will become weak leading to premature failure of the composite
3. In no case should be the second phase loose its strength, it should be well bonded to the matrix
4. Matrix must have a lower modulus of elasticity than the fibre.
5. In general, both the matrix and fiber should not have greatly different coefficient of linear expansion

Classification of composite materials based on reinforcement

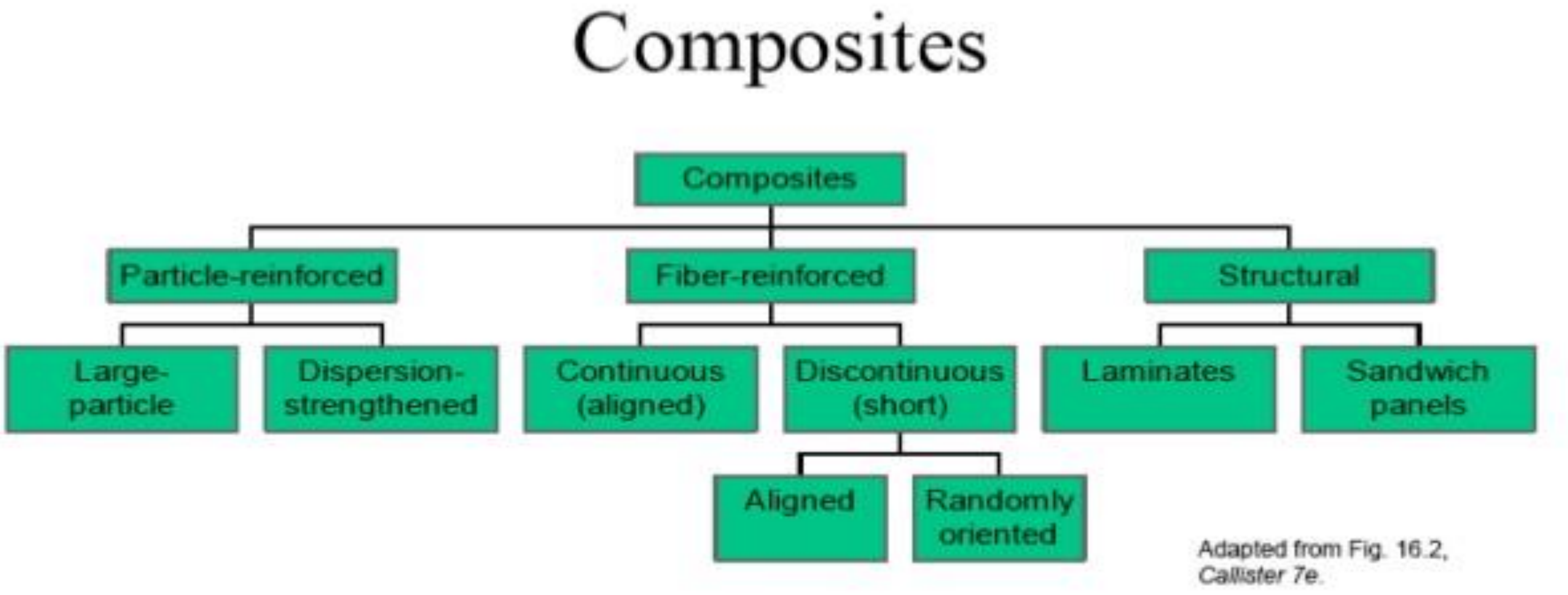


Fig. 2 Flow chart showing classification of composites

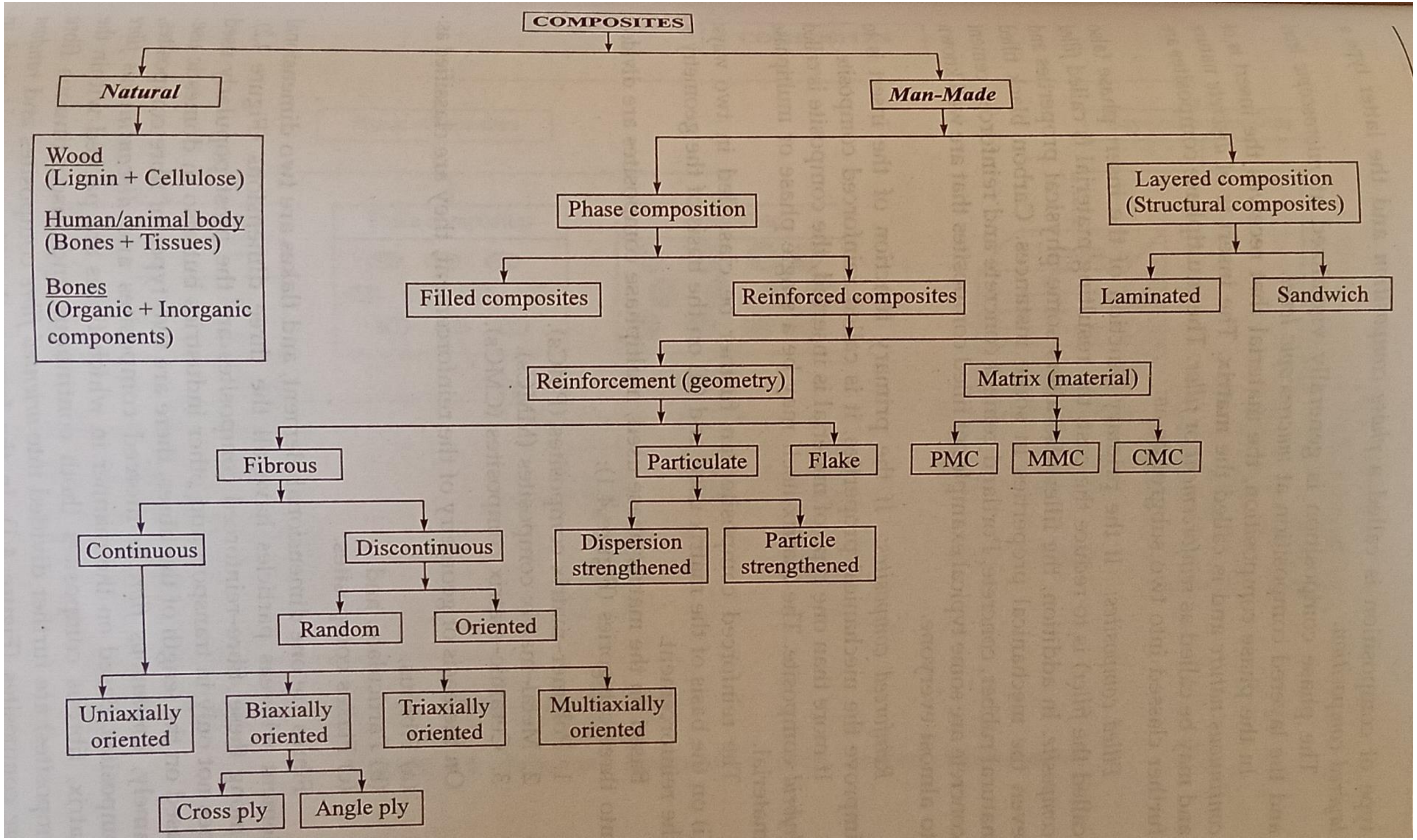


Fig. 3 Flow chart showing classification of composites

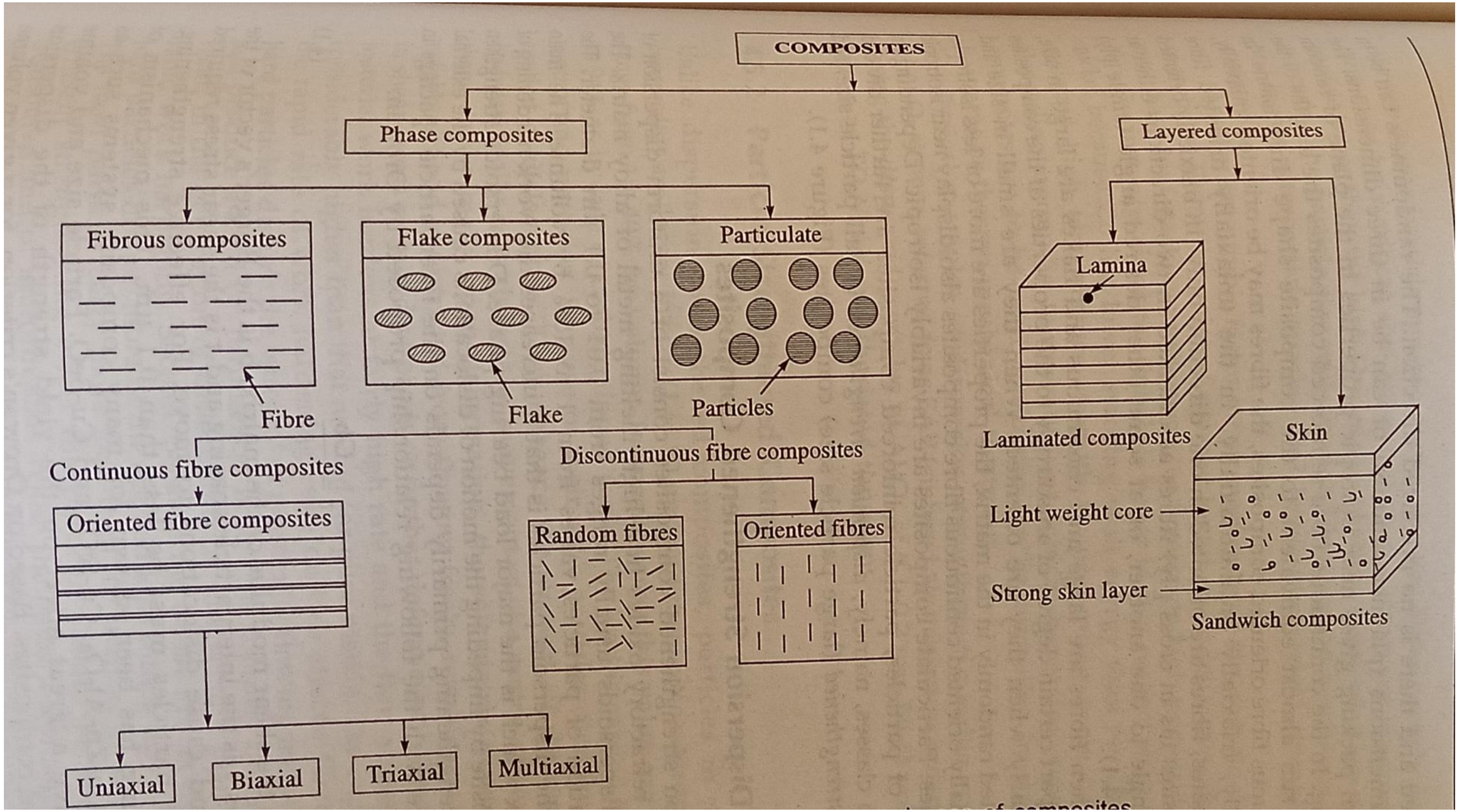


Fig. Schematic illustration of some classes of composites

In the phase composition, the material that receives the insert is of continuous nature and is called the matrix. The insert is of discrete nature and may be called as reinforcement or filler.

The multiphase composites are further classed into two subgroups as

(1) Filled Composites : If the primary function of the insert phase (also called the filler) is to reduce the cost, the resulting material is called filled composite. The filler also alters some physical properties and even the mechanical properties in some instances.

Ex. 1. Carbon black filled natural rubber

2. Concrete and Portland cement concrete and reinforced cement concrete are well known examples to everyone

Reinforced composites

If the primary function of the insert is to improve the mechanical properties, it is called reinforced composite

If more than one type of a material is inserted the composite is called hybrid composite. The matrix itself may be a single phase or multiphase material

The reinforced composite can further be classified in two ways

(i) On the basis of the matrix used

(ii) On the basis of the geometry of the reinforcement

On the basis of the matrix phase used, multiphase composites are divided into 3 categories

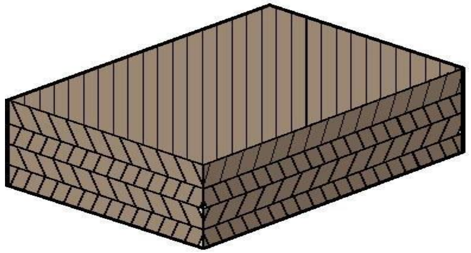
1. Polymer matrix composites (PMCs)
2. Metal-matrix composites (MMCs)
3. Ceramic-matrix composites (CMCs)

On the basis of geometry of the reinforcement, they are classified as

(a) Fibrous
(b) Particulate
(c) Flakes composites

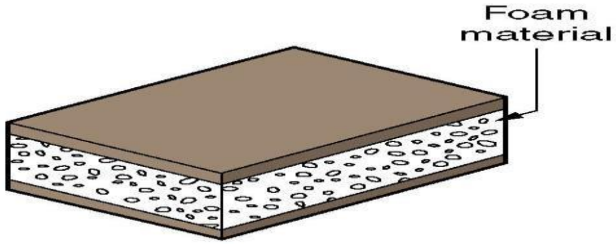
OTHER COMPOSITE STRUCTURES

Laminar composite structure :- *Two or more layers bonded together in an integral piece*

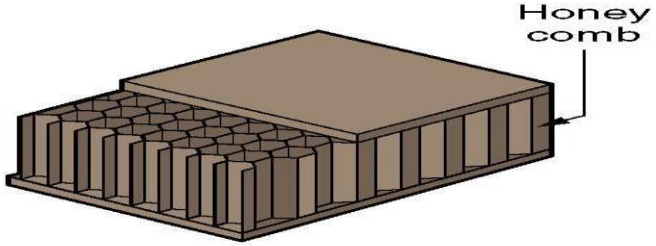


(a)

Sandwich structure:- *foam & honey cores*



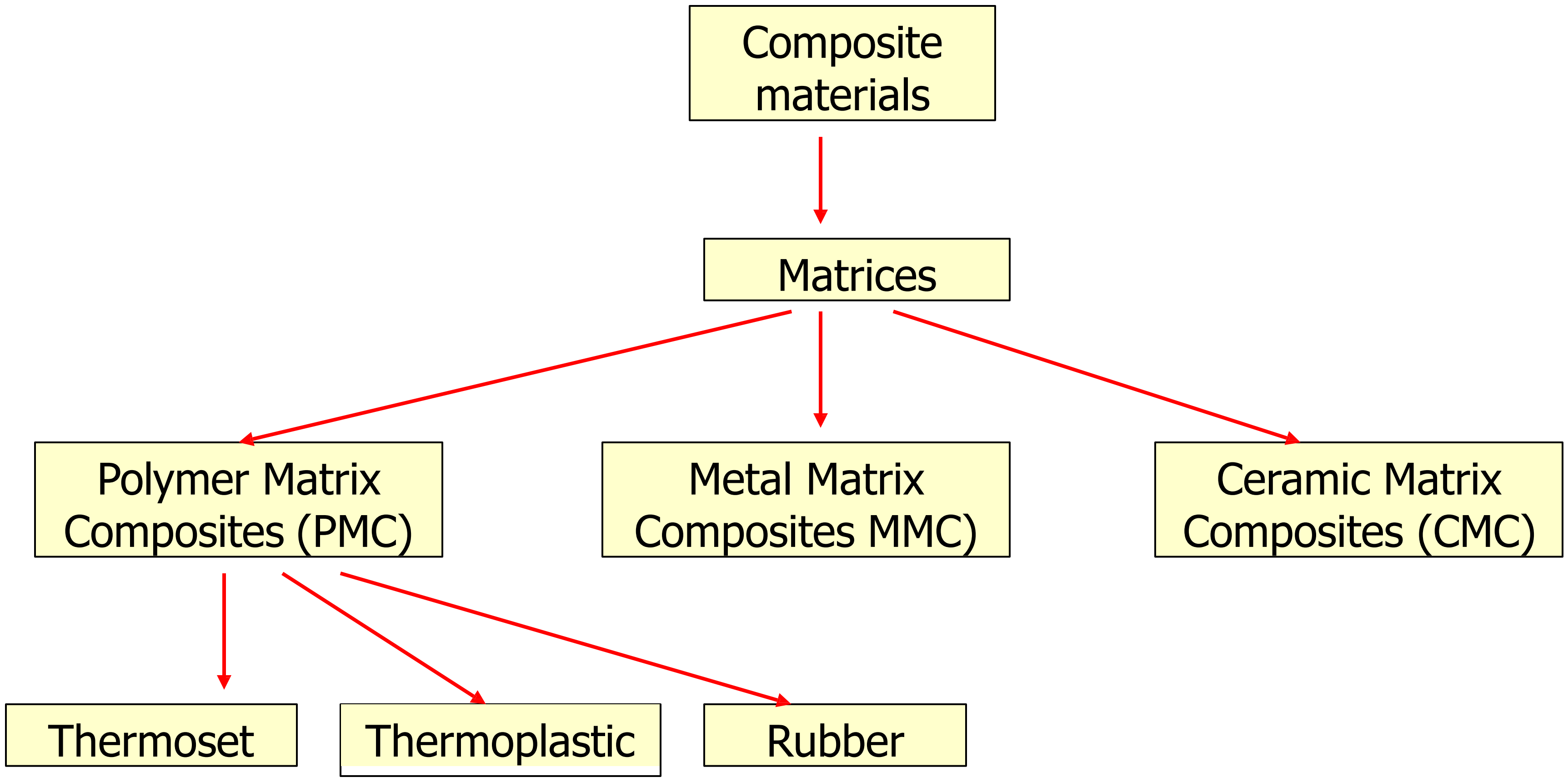
(b)



(c)

Consists of a thick core of low density foam bonded on both faces to thin sheets of a different material

Classification based on Matrices



Advantages of Composites

- Light in weight
- Strength-to-weight and Stiffness-to-weight are greater than steel or aluminum
- Fatigue properties are better than common engineering metals
- Composites cannot corrode like steel
- Possible to achieve combinations of properties not attainable with metals, ceramics, and polymers alone

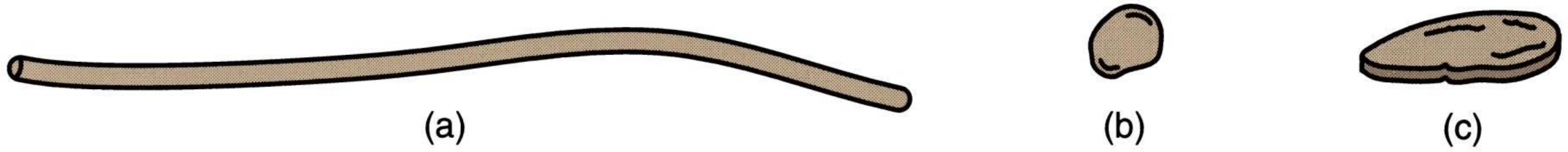
MATRIX & REINFORCEMENT

MATRIX

- Thermosetting polymers are the most common matrix materials
  - Principal TS polymers are:
    - Phenolics – *used with particulate reinforcing phases*
    - Polyesters and epoxies - *most closely associated with FRPs*
- Nearly all rubbers are reinforced with carbon black

REINFORCEMENT

- Possible geometries - (a) fibers, (b) particles & (c) flakes



- Particles and flakes are used in many plastic molding compounds
- Of most engineering interest is the use of fibers as the reinforcing phase in FRPs

Glass fibre reinforced polymer composites

- Contains glass fibre as reinforcing phase in a polymer matrix
- Glass fiber
  - Continuous or discontinuous
  - Dia b/w 3-20μm
- Lim: not very stiff and rigid
- Automobile and marine bodies, storage containers, plastic pipes, industrial flooring

Carbon fibre reinforced polymer composites

Contains carbon fibre as reinforcing phase in a polymer matrix

Advantages

- High specific strength and specific modulus
- Higher strength at elevated temperatures
- Not affected by moisture or acids

Applications

- Aerospace application
- Sports and recreational equipments
- Pressure vessels

Aramid fibre reinforced polymer composites

Aramid-polyamide-kevlar or nomex

- Advantages
  - High toughness and impact resistance
  - Resistance to creep and fatigue failure
- Applications
  - Bullet proof vests and armor
  - Sports goods, missile cases
  - Pressure vessels, clutch linings and gaskets

Metal matrix composites

Metals with low density and low temperature toughness are preferred as matrix metal.
Aluminium, titanium, magnesium and their alloys

- Advantages
  - Higher operating temperatures
  - Non flammability and creep resistance
  - Greater resistance to degradation by organic fluids.
- Applications
  - Aerospace application
  - Gas turbine blades
  - Electrical contacts

Ceramic Matrix Composites

- Primary objective for developing CMCs was to enhance the toughness while retaining the high temperature properties.
- High melting points and good resistance to oxidation
- But low tensile strength, impact resistant and shock resistance
- eg. Small particles of partially stabilized Zirconia are dispersed within a matrix material Al_2O_3

Hybrid composites

- Obtained by using different kinds of fibers in a single

Applications

1. Aircraft and aerospace applications
2. Automotive applications
3. Marine applications
4. Sporting industries
5. Biomaterials
6. Industrial applications

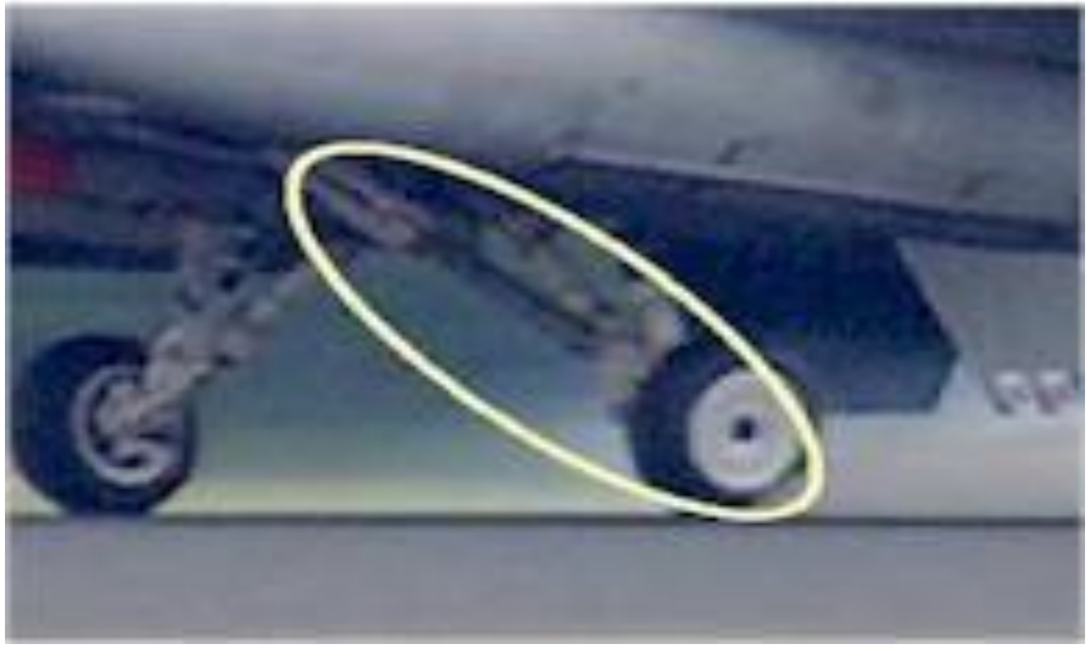
Lower Drag Brace for the F16

Lower Drag Brace of F16:

Landing gear

Titanium Matrix Composite

Monofilament SiC fibres in a Ti-matrix.



Brake rotors for high speed train

Brake rotors for high speed train

- particulate reinforced Aluminium alloy (AlSi7Mg+SiC particulates)
- Weight of cast iron rotor is 120 kg/piece while MMC it is 76 kg/piece

F-16 Ventral Fins

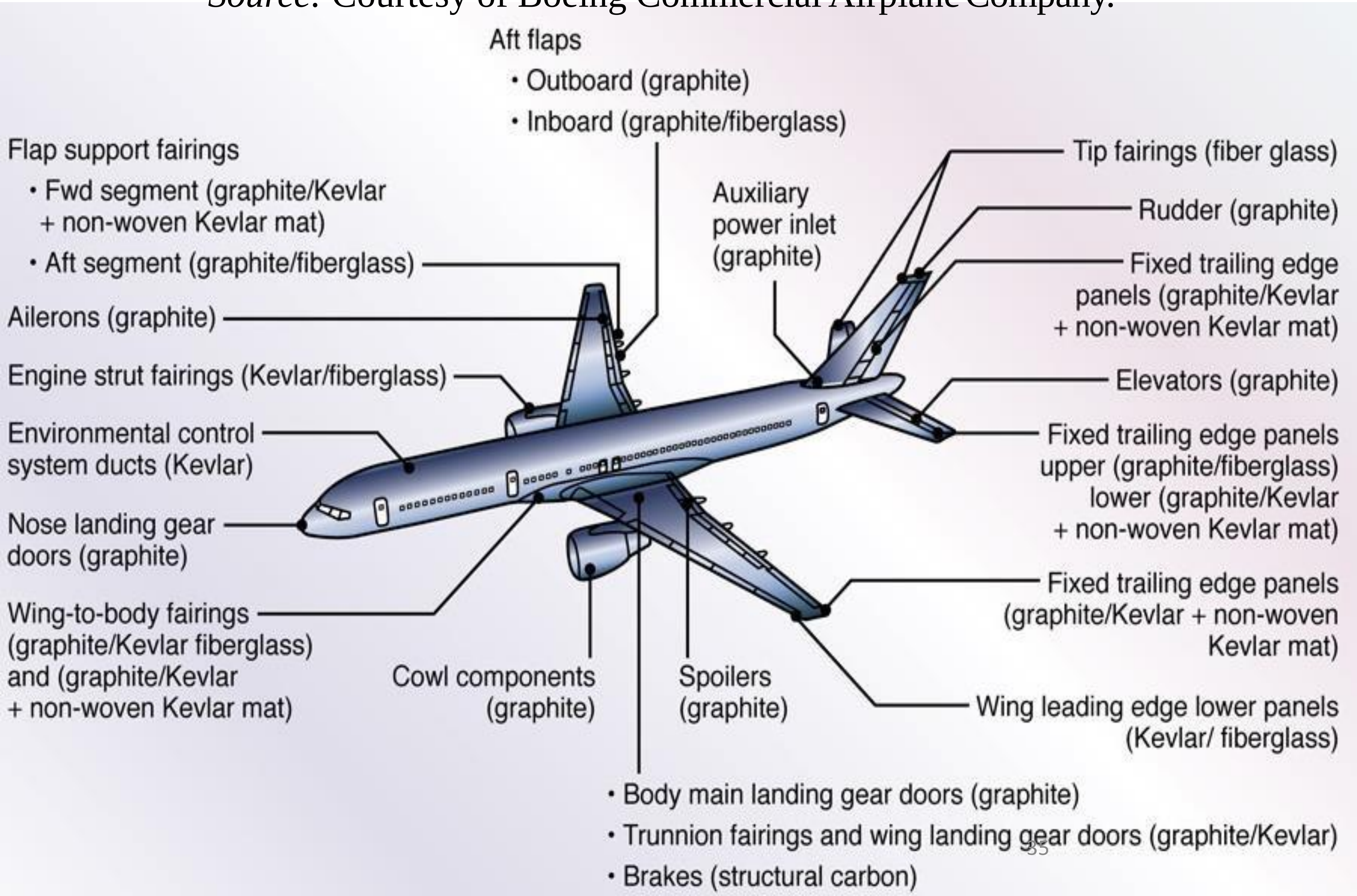
- High specific stiffness and strength

- Al-matrix composites
- Rolled P/M
- 6092/SiC/17.5p

- Decreased deflections provide an increase in fin life

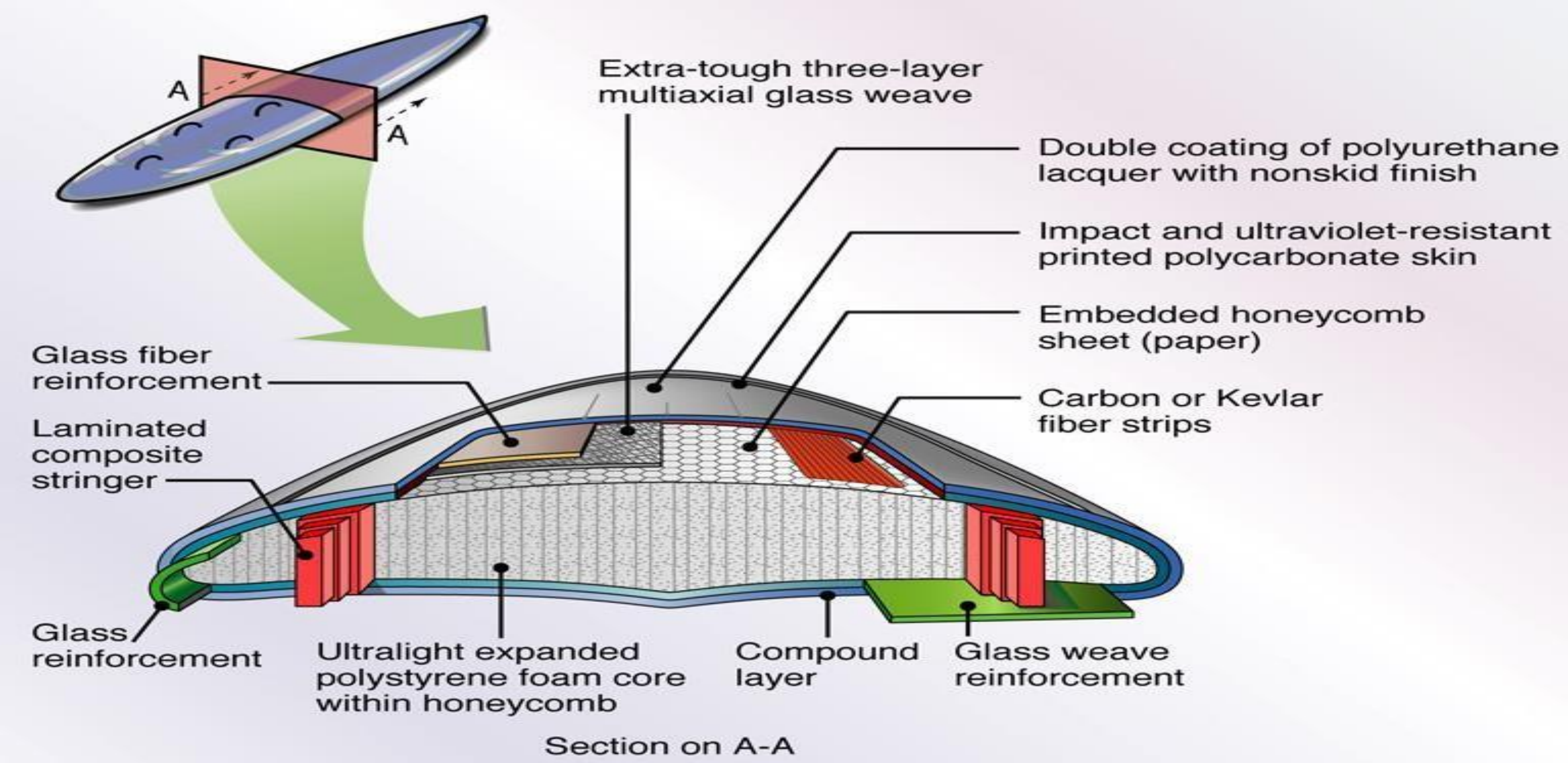
Application of advanced composite materials in Boeing 757-200 commercial aircraft.

Source: Courtesy of Boeing Commercial Airplane Company.

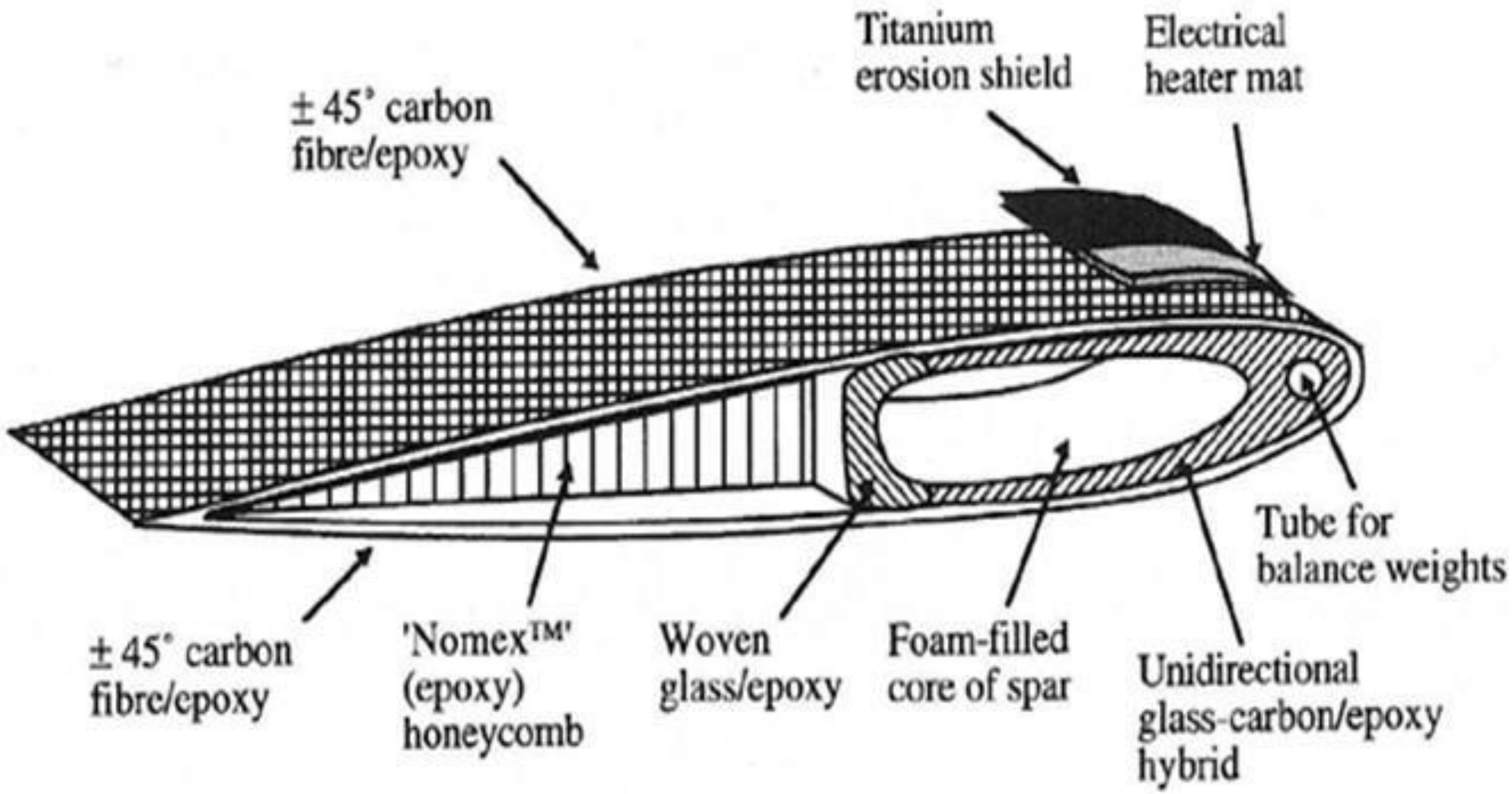


Cross-section of a composite sailboard.

Source: K. Easterline, *Tomorrow's Materials* (2nd ed.), p. 133. Institute of Metals, 1990.

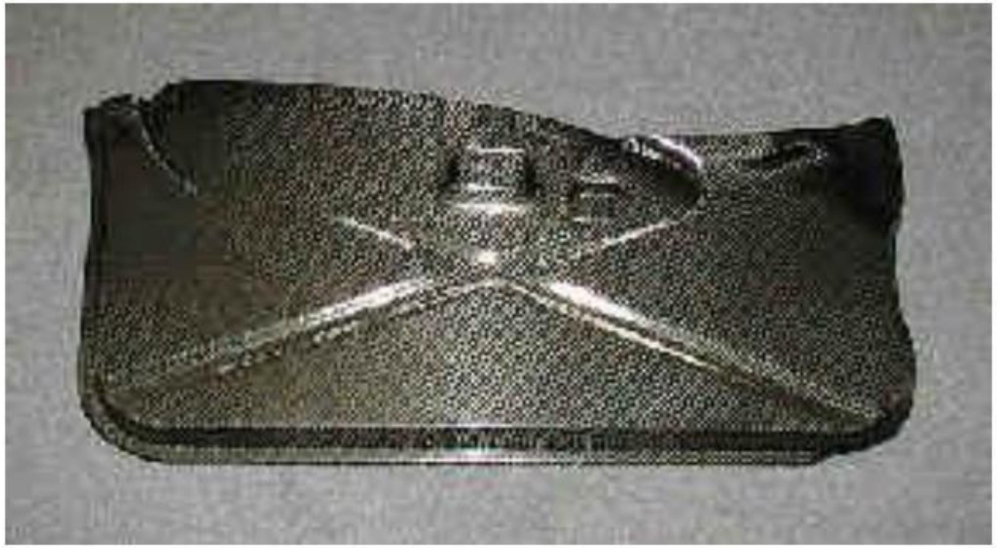


Helicopter rotor blade



Schematic section through a typical composite construction for a heli-
copter rotor blade. (Courtesy of Westland Helicopters.)

Wing Panel



Automotive structure

Aerospace parts

Sporting goods

Delta IV Rocket faring mandrel

Disadvantages and Limitations of Composite Materials

- Properties of many important composites are anisotropic - the properties differ depending on the direction in which they are measured – this may be an advantage or a disadvantage

- Many of the polymer-based composites are subject to attack by chemicals or solvents, just as the polymers themselves are susceptible to attack

- Composite materials are generally expensive

- Manufacturing methods for shaping composite materials are often slow and costly


THANK YOU